

Yanru Chen

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Research Focus

Computer architecture; hardware/software co-design; processing-in-memory and in-storage processing; heterogeneous memory; graph analytics; hyperdimensional computing; AI inference systems.

Education

University of California San Diego Sep 2024 - Present

Ph.D. candidate, Electrical and Computer Engineering | Anticipated graduation June 2027

Advisor: Tajana Rosing | GPA 3.6/4.0 | Qualifying exam passed March 2026

Tsinghua University Aug 2021 - Jun 2024

M.S., Electronic Information (Intelligent Manufacturing) | GPA 3.78/4.0

Jilin University Aug 2017 - Jun 2021

B.E., Electronic Science and Technology | College Honors | GPA 3.53/4.0 | Rank 6/108

Research Experience

Systems Energy Efficiency Lab | UC San Diego Feb 2025 - Present

Graduate Researcher | Advisor: Tajana Rosing

- Research in computer architecture, hardware/software co-design, processing-in-memory, in-storage processing, graph analytics, hyperdimensional computing and databases.
- Shared-first-author work on CHIME and RAPID-Graph (DATE 2026) and HDDB (DAC 2026); co-author of PALUTE (ISLPED 2026); first-author GEN-Graph manuscript submitted to IEEE TCAD.

Flexible and Printed Electronics Lab | UC San Diego Sep 2024 - Jan 2025

- Studied memristor fabrication and a multimodal sensing glove, with a focus on MEMS/NEMS fabrication and biomedical signal processing.

Micro Electro Mechanical System Laboratory | Tsinghua University Oct 2021 - Jun 2024

- Wearable acetone gas sensor for breath detection and analysis: wearable sensors, semiconductor gas sensors and microfabrication (Oct 2021 - Jun 2024).
- Studied laser-induced graphene flexible strain sensors and associated microfabrication techniques (Sep 2022 - Nov 2023).

Computational Medicine Laboratory | Western University Jan - Aug 2020

- CT perfusion heterogeneity analysis using PM3 and machine learning; fractal-dimension analysis and reconstruction of 3D perfusion maps.

Industry Experience

SK hynix | San Jose, CA Jun - Sep 2026

Emerging Memory AI Infrastructure Intern

- Built event-driven simulation and trace-replay workflows to evaluate heterogeneous AI architectures and inform hardware/software co-design.

Shenzhen Hanit Industrial Technologies Co. | Shenzhen, China Jun - Oct 2023

Assistant Engineer, Department of Hardware

- Developed a flexible strain sensor project using LiG-SiC nanosheets with self-temperature compensation.

Vivolight Medical Device and Technology Co. | Shenzhen, China Apr - May 2021

Algorithm Intern, Department of Algorithm

- Worked on OCT imaging integrating near-infrared spectroscopy with a machine-learning-based lipid imaging system.

Conference Publications

* Equal contribution.

- [1] Runyang Tian, **Yanru Chen**, Weihong Xu, and Tajana Rosing. PALUTE: Processing-In-Memory Acceleration via Lookup Table for Edge LLM Inference. *IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED)*, 2026.
- [2] Quanling Zhao*, **Yanru Chen***, Runyang Tian, Sumukh Pingre, Weihong Xu, Augusto Vega, Steven Holmes, Saransh Gupta, and Tajana Rosing. HDDB: Efficient In-Storage SQL Database Search Using Hyperdimensional Computing on Ferroelectric NAND Flash. *The Chips to Systems Conference (DAC)*, 2026.
- [3] **Yanru Chen***, Zheyu Li*, Keming Fan, Runyang Tian, John Hsu, Weihong Xu, Minxuan Zhou, and Tajana Rosing. RAPID-Graph: Recursive All-Pairs Shortest Paths Using Processing-in-Memory for Dynamic Programming on Graphs. *Design, Automation and Test in Europe Conference (DATE)*, 2026.
- [4] **Yanru Chen***, Runyang Tian*, Yue Pan, Zheyu Li, Weihong Xu, and Tajana Rosing. CHIME: Chiplet-based Heterogeneous Near-Memory Acceleration for Edge Multimodal LLM Inference. *Design, Automation and Test in Europe Conference (DATE)*, 2026.
- [5] **Yanru Chen**, Jiaqi Liu, Yixin Liu, and Min Zhang. Enhanced Strain Resistance of Fractal Fiber Laser-Induced Graphene for Flexible Electrodes Via Annealing and Plasma Etching. *23rd International Conference on Solid-State Sensors, Actuators and Microsystems (Transducers)*, 2025, pp. 1109-1112. Poster presented by Yanru Chen.
- [6] Yixin Liu*, **Yanru Chen***, Zhibiao Wang, and Min Zhang. Enhanced Electrical Conductivity in Laser-Induced Graphene-Silicon Carbide Laminated Nanosheets for Flexible Strain Sensors and Pulse Wave Velocity Assessment. *IEEE 37th International Conference on Micro Electro Mechanical Systems (MEMS)*, Austin, TX, USA, 2024, pp. 919-922. Poster presented by Yanru Chen.
- [7] **Yanru Chen**, Sanjay R. Kharche, Yan Min Zhang, G. H. Janssen, and Christopher W. McIntyre. Quantifying microvascular alterations due to a pharmacological agent. *42nd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Montreal, QC, Canada, 2020. Oral presentation by Yanru Chen.
- [8] Sanjay R. Kharche, **Yanru Chen**, and Christopher W. McIntyre. Fractal Dimension Based Texture Analysis of CT Perfusion Imaging. *42nd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Montreal, QC, Canada, 2020. Oral presentation by Yanru Chen.

Journal Publications

[1] **Yanru Chen**, Yixin Liu, Jiaqi Liu, Yuzhen Li, Yuhan Liu, Wenjie Zhang, Liuyang Han, Dongkai Wang, Shuhong Cao, Hanxiao Liu, and Min Zhang. Porous PDMS-ZnO Wearable Gas Sensor for Acetone Biomarker Detection and Breath Analysis. *ACS Applied Materials & Interfaces*, 2024.

[2] Jiaqi Liu, Yuan Yu, Rujun Zhang, **Yanru Chen**, Yanlong Guo, Yi Zhang, Ran Tao, Jing-Ting Luo, Hairong Zheng, Pingfa Feng, and Min Zhang. Trapping nanoscale particles via quasi-Scholte mode in acoustofluidics. *Lab on a Chip*, 2025.

[3] Yuhan Liu, Liuyang Han, Siqi Lv, Tao Jiang, Mingkai Duan, Hanyu Guo, Yuzhen Li, Qisen Xie, **Yanru Chen**, Dongkai Wang, and Min Zhang. Emotional and Directional Enabled Programmable Flexible Haptic Interface for Enhanced Cognition in Disabled Community. *Research*, 2025.

[4] Qisen Xie, Liuyang Han, Jie Liu, Wenjie Zhang, Liuyan Zhao, Yuhan Liu, **Yanru Chen**, Yuzhen Li, Qian Zhou, Ying Dong, and Min Zhang. Kirigami-Inspired Stretchable Piezoelectret Sensor for Analysis and Assessment of Parkinson's Tremor. *Advanced Healthcare Materials*, 2025.

[5] Yuzhen Li, Yunfei Wang, Yixin Liu, Yuhan Liu, Liuyang Han, **Yanru Chen**, Juntian Qu, Puxiang Lai, and Xiang Qian. Simultaneously encapsulation and formation of PDMS-MWCNTs composites for multi-directional microchannel force sensors. *IEEE Sensors Journal*, 2024.

Submitted Manuscripts and Preprints

[1] **Yanru Chen**, Runyang Tian, Zheyu Li, Mahbod Afarin, Weihong Xu, and Tajana Rosing. GEN-Graph: Heterogeneous PIM Accelerator for General Computational Patterns in Graph-based Dynamic Programming. *Submitted to IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*.

[2] **Yanru Chen**. Low-Cost Spin-Coating Process for SnO₂-Based Memristors: Comparative Study of Different Solvent Systems and Device Performance. *SSRN 5208743*, 2025. *Preprint*.

Awards and Scholarships

UCSD CSE Best Doctoral Research Award | Jun 2026

DAC Young Fellow Award | Mar 2026

Workshop on Learning and Information Theory, Second Best Oral Presentation Award | Nov 2024

Jacobs Fellowship, University of California San Diego | Aug 2024

Outstanding Undergraduate Graduation Project Award, Jilin University | Jun 2021

Excellent Student of the College (Top 5%), Jilin University | 2018 - 2021

MITACS Globalink Research Internship Award, Western University | Feb 2020

Technical Skills

Programming: Python, C++, MATLAB, Verilog

EDA and simulation tools: Altium Designer, Cadence Virtuoso, COMSOL, AutoCAD

Platforms and microcontrollers: Arduino, ESP32, FPGA (Xilinx/Altera)